

Mark Samuel

Richmond Hill, Ontario, Canada — Available Winter & Summer 2027

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Education

University of Toronto

Expected May 2028

B.A.Sc. in Electrical Engineering - Deans List — Minors: AI/ML & Robotics/Mechatronics

Toronto, ON

- Relevant coursework: Computer Architecture, Digital Electronics, Hardware Design, Computer Hardware, Data Structures & Algorithms, Machine Learning, Artificial Intelligence, C/C++ Programming, Biomedical Imaging.

Technical Skills

Integration & Implementation: RTL lint (SpyGlass/Leda), CDC (Questa CDC), simulation (VCS, ModelSim), logic synthesis (Design Compiler), netlist review, SRAM/memory-compiler flows, ECC wrappers, PPA analysis, LSF.

Digital Design: Verilog, FSMs, pipelining, memory hierarchy, CDC synchronizers, async FIFOs, PCIe (TL/PL), RISC-V ISA/Assembly, FPGA (Quartus, DE1-SoC), Transistor-level CMOS schematic and layout (SUE, MAX).

Embedded, Analog & Lab: C, C++, STM32, ESP32, Arduino, PWM, I2C, SPI, UART, BLE HID, HW/SW co-design; LTspice, Altium, 2-layer PCB layout, soldering and rework, PCB bring-up and debug; oscilloscope, multimeter, function generator.

Software & Tooling: Python, Tcl, Bash, SQL, JavaScript, MATLAB; Linux, Perforce, Git; React, FastAPI, Postgres, HTML/CSS.

Experience

Silicon Implementation & Integration Intern — PCIe Subsystem

May 2026 – Present

AMD (Advanced Micro Devices)

Markham, ON

- Own the team's Tcl async-FIFO extraction script, combining RTL and build-time feature-flag analysis to classify edge cases and generate changelist-pinned reference lists that partner teams use for post-synthesis equivalence, read/write-pointer race, and gray-code crossing checks; co-developed an LLM-based agent skill automating RTL-to-netlist LEC in Cadence Conformal.
- Triage RTL lint and CDC violations at the BIF core and subsystem level, driving root cause with designers and often proposing the fix; drove design waivers toward zero by pushing owners for a fix or documented sign-off.
- Analyze synthesis QoR across releases (cell count, std-cell, hard-macro and RAM area, ULVT VT mix, levels of logic and max flop-to-flop depth) and compare SRAM genIP lists across memory releases, flagging PPA regressions and bbox area bloat to designers.
- Built an automated PCIe connectivity visualizer (FastAPI, React Flow) that traces signals from top-level Verilog through every module instantiation to diagram 12+ tiles, showing bundle widths color-coded by repeater constraint (EQ0, LE1), across 8+ projects.

Team Leader — Battle Bots Design Team

Sep 2024 – Mar 2025

University of Toronto Robotics Association

Toronto, ON

- Led a team of 6 to 1st place (20+ teams) at UoT's Battle Bots Competition; designed a robust 3D-modeled chassis in AutoCAD, integrating IR and ultrasonic sensors for opponent tracking.
- Programmed hardware interrupts in Arduino C++ for deterministic sensor response, reducing detection error by 60%.
- Implemented PWM motor drivers and debugged high-speed spin stability, synchronizing actuator response with control logic across the hardware/software boundary under real-time constraints.

Projects

Embedded Gesture-Based Presentation Controller | STM32, ESP32, BLE HID, C

Mar 2026

- **Placed 3rd of 100+ course teams** with a wireless gesture controller built on an STM32F446ZE, an IMU, and an ESP32 BLE interface. Mapped IMU quaternion data to cursor motion and slide-change gestures over a UART MCU-to-MCU link. Analyzed end-to-end latency, bandwidth, and power tradeoffs across the sensing, firmware, wireless, and feedback systems.

Power Amplifier and Low Pass Filter PCB | LTSpice, Altium, PCB Assembly, Python

Jan 2025

- Designed a 40 V_{pp} Class D power amplifier with a 16 MHz LPF; compared BJT, comparator, and MOSFET topologies in LTSpice to handle high currents at elevated frequencies. Fabricated and tested a PCB, soldering and assembling RF-suitable components; validated performance using bench instruments and developed Python scripts for automated debugging.

RISC-V Soft-CPU Physics Visualizer | C, RISC-V Soft CPU, DE1-SoC FPGA, VGA, ModelSim

Mar 2025

- Deployed a RISC-V soft CPU on a DE1-SoC FPGA running particle-dynamics simulations written in C with VGA output. Met FPGA timing/memory constraints by optimizing computations with Taylor-series approximations. Validated behavior in ModelSim and on-board, tracing C-level performance down to RISC-V assembly, pipeline stalls, and memory limits.

Piano Tiles Game in Verilog | Quartus Prime, ModelSim, VGA, FSM Design

Dec 2024

- Implemented a Piano Tiles game in Verilog with keyboard input, speaker output, and VGA rendering using ROM for tile graphics and memory-mapped I/O for peripherals. Designed control logic through a finite state machine (FSM) for game progression, validated correctness and timing via ModelSim simulation, then deployed on FPGA hardware.